Vipin Neekamparambath

Agentic AI Engineer | Senior Analyst, Accenture DACH | Engineering Digitization

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Professional Summary

Agentic AI Engineer and Senior Analyst at Accenture DACH with 10+ years across CAE automation, engineering digitization, and enterprise delivery. I architect multi-agent workflows combining RAG, tools, validation, and human review across PLM, requirements, quality, and simulation. Hands-on with LangChain, LangGraph, Deep Agents, MCP, Python, Azure, and cross-functional delivery.

Evidence at a Glance

10+ years in engineering automation and digital delivery | **5** Agentic AI and GenAI case studies | **~90%** smaller Abaqus ODB files in a delivered Python utility | **4** core workflows: PLM, requirements, quality, simulation

Professional Experience

Accenture DACH - Senior Analyst

Munich, Germany | Aug 2023 - Present

- Architect an Engineering Orchestrator coordinating agents across requirements, PLM, design, simulation, and downstream lifecycle workflows.
- Serve as development lead for end-to-end FMEA agents, shaping workflow decomposition, agent responsibilities, prompts, tools, and validation.
- Lead requirements engineering and requirements management agents using the Deep Agents framework.
- Develop retrieval and knowledge-management capabilities connecting critical engineering data with reusable process guidance.
- **Outcome:** Connected previously separate engineering activities through reusable agent, tool, and knowledge patterns.

Altair Engineering - Technical Specialist

Bengaluru, India | Jan 2022 - Jul 2023

- Designed, tested, and deployed tools for engineering pre-processing, post-processing, and report generation.
- Delivered end-to-end CAE automation that reduced repetitive engineering effort and improved workflow consistency.
- **Outcome:** Reduced repetitive engineering work while making delivery more consistent.

ZF Friedrichshafen - Specialist

Hyderabad, India | Jun 2021 - Jan 2022

- Managed a centralized Azure Repos library for engineering macro collections.
- Built a Microsoft Power Apps script-library application for cross-functional teams.
- **Outcome:** Improved access to reusable engineering scripts across cross-functional teams.

Valeo - Automation Engineer

Chennai, India | Jan 2018 - Jun 2021

- Established work instructions, GUI and script templates, quality checklists, and team standards.
- Built dashboards to track specifications, delivery progress, and engineering time savings.
- **Outcome:** Created the operating standards and visibility needed for a new automation capability.

Tata Consultancy Services - Systems Engineer

Bengaluru, India | Nov 2014 - Dec 2017

- Created HyperWorks scripts that improved FE modelling efficiency and reinforced CAE quality standards.
- Supported vehicle subsystem assembly in HyperMesh and engineering revisions in UG NX.
- **Outcome:** Built first-hand understanding of the engineering work later targeted by automation and AI.

Selected Agentic AI and Automation Work

Engineering Orchestrator | Agentic AI | LangChain | Azure
A multi-agent operating layer connecting requirements, PLM, design, simulation, and downstream lifecycle workflows. Impact: Created a cohesive AI-assisted layer across previously separate lifecycle activities.

FMEA Analysis Agents | LangGraph | LLM orchestration | Human review
End-to-end agents supporting workflow decomposition, context retrieval, analysis guidance, validation, and structured FMEA outputs. Impact: Made an expert workflow more consistent and easier to inspect.

Requirements Engineering Agents | Deep Agents | Requirements engineering | Tool integration
Agents assisting requirements elicitation, analysis, interpretation, management, and traceability-oriented work. Impact: Improved access to process guidance during requirements work.

Abaqus ODB Compression Tool | Abaqus | Python | Simulation data
A standalone Python utility that reduced large Abaqus ODB result files by approximately 90% through selective variable and frame extraction. Impact: Achieved an approximately 90% compression ratio on large ODB files.

Additional case studies and evidence trails: cv-vipin.vercel.app

Capabilities

Agent architecture: Multi-agent orchestration | LangChain | LangGraph | Deep Agents | Model Context Protocol (MCP) | Google Agent Development Kit (ADK)

Knowledge and GenAI: RAG and knowledge retrieval | Vector databases for RAG | Prompt and workflow design | AI copilots | Large language models (LLMs)

Engineering delivery: Requirements engineering | FMEA and quality | CAE and FEM automation | Python and Tcl/Tk | Engineering report automation

Cloud and delivery platforms: Azure AI Foundry | Azure DevOps | Azure Portal | Google Cloud Platform (GCP)

Delivery and leadership: Agile delivery | Project planning | Stakeholder management | Cross-functional collaboration | Technical leadership | Training delivery

Education and Languages

Master of Technology, Industrial Engineering & Management | National Institute of Technology Calicut | 2012 - 2014
Publication: Optimization of assembly job-shop scheduling using priority dispatching rules | CGPA: 7.4/10

Languages: English | Hindi | German - A1